

Q1: what are the different protocols you observed at the following layer of protocol stack?

Soln As per instruction given, I captured 3510 packets.

① Application layer

In these packets, we get these application layer protocol i.e :-

- i) DNS → Displayed (274), about (7.8%)
- ii) HTTP → Displayed (8), about (0.2%)
- ~~iii) HTTPS →~~
- iii) DHCP → Displayed (11), about (0.3%)
- iv) SSDP → Displayed (28), about (0.7%)

② Transport layer

In these packets, we get these transport layer Protocol

- (i) TCP → Displayed (1029), about (29.3%)
- (ii) UDP → Displayed (1106), about (31.5%)

③ Network layer

- i) ICMPV6 → Displayed (1), about (0.0%)
- ii) IGMP → Displayed (2), about (0.1%)
- (iii) IPV6 → Displayed (136), about (3.9%)
- iv) ARP → Displayed (1335), about (38.0%)

(2)

Q.2 what is the total amount of data being received for the following two case?

Soln :- ① Neverssl.com $\rightarrow$ [IP. address = 34.223.124.45]

Total amount of Packet received = 189 of 2133
with 825 Bytes of Data. of 6936 bits about (8.9%)

② Cornell.edu : [IP. Address = 128.253.173.241]

Total Packet of 823 bytes found. of 6920 bits

Q.3 Answer the following when you access the site neverssl.com $\rightarrow$

Soln 1. How many HTTP GET request do you observe?
List down the GET request.

Soln There are 6 HTTP GET request; observed

i) 378 GET / HTTP/1.1

ii) 367 GET / canonical.html HTTP/1.1

iii) 384 GET / success.txt? IPv4 HTTP/1.1

iv) 465 GET / online HTTP/1.1

v) 466 GET / online / HTTP/1.1

vi) 472 GET / favicon.ico HTTP/1.1

2. what version of HTTP the Server running?

Sol $\rightarrow$ only HTTP/1.1

3) For each of the HTTP GET requests, find out

(3)

① The total no. of TCP segment being received.

Sol 2133 TCP Segment being received.

② The Total amount of data being : received in the corresponding HTTP: request message.

Sol Total amount of data being received in the corresponding HTTP request message is 608.

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4. Identify TCP three way handshake used to establish the connection with the netvsnl.com. Specify SYN, SYN-ACK, ACK packets & write the sequence of numbers of each of the three accesses.

Soln Get Request

No →

→ GET/online HTTP/1.1

Stream No: 15

No.	Info	Sequence No.
612	[SYN]	0
1458	[SYN, ACK]	0
1459	[ACK]	1